10.1.3 Service level objective

SLOs are system-external goals, and you can use one or more of them in the context of informing SLAs. Having multiple SLOs for a service enables you to combine them for a comprehensive pass/fail test. In contrast to SLAs, the audience for SLOs is not your service customer; it's your service team. You, as a product owner, define SLOs together with your engineers in cooperation with sales to determine which errors are acceptable to the users of a service. For example, you may agree that for a specific service, you want to reach 90% availability over the time period of a month. Then, at the end of the month, you may compare the SLIs with the SLOs to determine whether you met the goal.

The balancing act of setting SLO thresholds is made between the competing demands of availability and velocity. Think of the extreme case of an SLO that states 0% availability. This is easy to achieve; the service can be down any time and you can ship new code into production at will. That's great from a developer's perspective, since developers can push new features or bug fixes at any time. From a service customer point of view, it is less great; it means they cannot rely on the service at all. The other end of the spectrum would be 100% availability, meaning that, effectively, no changes are allowed and no bugs or other factors bring down your service. Clearly, neither extreme is desirable or realistic.

> **NOTE** In some respects, SLOs are a bit like the speed of light: the closer you want to get to it, the more energy you have to invest. Setting an SLO of 99% versus 99.9% may only look like 0.9% more, from a goal perspective, but consider the work that has to go into achieving it. Also, remember that the weakest link in the request path determines what you can achieve, overall. For example, if you have three services—A, B, and C—where A is calling B; B is calling C; and, let's say, A and B have an SLO of 99.9% and C has an SLO of 90%, you can not expect to promise 99.9% overall, since C will not support this. See Steve McGhee's SLOconf 2021 talk, "SLO Math," (https://www.youtube.com/watch?v=-lHPDx90Ppg) for more information.

There is more terminology related to SLOs, such as the *error budget*, which defines how much wiggle room you have left for an SLO with respect to the SLIs in a time period, and *burn rate*, which defines how quickly you are using up your error budget. While the terminology can be a bit dense, in a nutshell, error budgets allow you to significantly improve alert accuracy, reduce noisy alerts, and improve practice leading indicators. I was told by Nobl9 representatives that there are case studies with customers that show a 90%+ reduction in false positive alerts after implementing SLOs.

We will not go into further details here. If you are interested in learning more about the advanced terminology (which is not widely adopted outside of Google), check out the freely available SRE books (https://sre.google/books/). To learn more about where SLOs originally stem from, I recommend "The Origin of Service Level Objectives" by Akshay Chugh and Piyush Verma (http://mng.bz/gBMe).